

DUCK PIER, NC, USA

WMO: 998209

Lat: 36.184N

Lon: 75.746W

Elev: 23

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.4	(c) 28.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 38.2	(k) 42.9	(l) 35.1	(m) 37.7	(n) 18.7	(o) 330	(p) 0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.8	(c) 88.5	(d) N/A	(e) 86.3	(f) N/A	(g) 84.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 12.4	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 36.5	(b) 31.7	(c) 27.3	(e) 20.1	(f) 92.4	(g) 4.6	(h) 2.1	(i) 16.8	(j) 93.9	(k) 14.1	(l) 95.2	(m) 11.5	(n) 96.3	(o) 8.2	(p) 97.8		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	42.8	44.4	49.9	58.4	67.0	75.6	78.9	78.4	74.3	66.1	55.9	48.9		
	DBStd	14.47	9.69	7.67	7.77	7.45	6.80	5.13	3.64	3.28	3.92	6.41	6.74	8.18		
	HDD50	716	267	188	104	13	0	0	0	0	0	1	23	120		
	HDD65	2866	689	576	470	222	60	2	0	0	0	64	281	502		
	CDD50	5029	42	33	101	266	527	768	897	881	730	500	199	85		
	CDD65	1704	0	1	2	25	122	320	432	416	280	97	7	2		
	CDH74	11103	0	0	1	63	458	2104	3754	3310	1224	186	3	0		
	CDH80	2109	0	0	0	1	66	469	857	589	118	9	0	0		
(14) Wind	WSAvg	13.9	14.7	14.5	15.9	14.7	14.5	13.0	11.5	11.4	13.9	14.1	14.6	14.0		
(15) Precipitation	PrecAvg	55.3	4.3	3.7	4.4	3.7	4.3	5.0	6.2	6.0	5.7	4.2	3.5	4.2		
	PrecMax	73.7	7.5	7.8	8.8	7.0	8.0	9.4	10.0	11.5	19.9	8.7	9.3	8.6		
	PrecMin	38.5	1.8	1.4	1.3	1.3	1.7	1.8	3.7	2.4	1.4	0.1	0.8	1.7		
	PrecStd	7.4	1.4	1.6	1.5	1.4	1.7	1.9	1.5	2.1	3.9	1.9	2.0	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.0	68.8	72.7	78.9	85.7	90.7	92.0	90.5	86.1	80.8	73.4	71.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	64.8	64.9	68.6	75.9	81.6	87.5	88.9	87.0	83.2	77.9	70.5	67.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	61.6	61.6	65.8	73.1	78.9	85.1	86.4	84.9	81.0	76.1	67.9	64.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	57.3	56.6	62.6	70.1	76.3	82.5	84.3	83.1	79.4	74.4	65.3	61.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	10.6	11.2	11.0	9.6	8.3	7.8	6.6	5.4	7.3	8.8	9.7		
		MCDBR	15.3	16.7	16.9	16.1	13.4	11.9	11.0	9.9	8.6	8.8	11.7	13.4		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.329	0.353	0.396	0.440	0.486	0.514	0.502	0.416	0.366	0.333	0.325		
	taud		2.537	2.508	2.467	2.356	2.271	2.193	2.123	2.145	2.387	2.511	2.546	2.551		
	Ebn at Noon		279	289	291	281	269	256	248	248	266	272	272	270		
	Edh at Noon		25	29	33	39	43	46	49	47	36	29	25	23		
(44) All-Sky Solar Radiation	RadAvg		745	994	1338	1710	1925	2041	1933	1752	1448	1156	843	673		
	RadStd		55	100	101	121	141	115	96	115	97	115	62	52		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+203

Nomenclature: See separate page